

RFID-Controlled Solenoid Access Circuit Package (Revised)

This revision replaces the earlier block-style drawing with a true schematic-style circuit document showing explicit component symbols, pin-level IC/module representations, resistor dividers for RFID logic-level reduction, the MOSFET lock driver, and support parts.

Design summary

An Arduino Nano reads an MFRC522 module over SPI. The MFRC522 is powered from a dedicated 3.3 V regulator. Four Nano outputs that drive MFRC522 inputs are reduced from 5 V to about 3.0 V using resistor dividers. When a scanned UID matches the programmed allowlist, the Nano drives a logic-level MOSFET that energizes a 12 V cabinet lock solenoid. A flyback diode suppresses inductive kickback. Two LEDs and an optional buzzer provide operator feedback.

Schematic files

SVG schematic source: /workspace/rfid_solenoid_access_v2/schematic_v2.svg
Python generator: /workspace/rfid_solenoid_access_v2/generate_schematic.py
The schematic is delivered as SVG because a direct SVG->PNG conversion tool was not present in the environment. The SVG itself is the primary final schematic artifact.

BOM summary

Item	Refs	Qty	Description	Unit \$	Ext \$
1	U1	1	Arduino Nano	27.1	27.1
2	U2	1	MFRC522 RFID module	9.9	9.9
3	REG1	1	3.3V regulator	3.15	3.15
4	Q1	1	Logic-level N-MOSFET	2.5	2.5
5	D1	1	Flyback diode	0.54	0.54
6	LOCK1	1	12V cabinet solenoid lock	8.99	8.99
7	R1	1	Gate resistor	0.1	0.1
8	R2	1	Gate pulldown	0.1	0.1
9	R3,R4	2	LED resistors	0.1	0.2
10	R5,R6,R7,R11	4	Top divider resistors	0.1	0.4
11	R8,R9,R10,R12	4	Bottom divider resistors	0.1	0.4
12	LED1	1	Green LED	0.2	0.2
13	LED2	1	Red LED	0.2	0.2
14	BZ1	1	Active buzzer	1.0	1.0
15	C1	1	Bulk capacitor	0.35	0.35
16	C2	1	Regulator output capacitor	0.2	0.2

Total estimated cost: \$55.33 USD

Firmware

Arduino sketch: `/workspace/rfid_solenoid_access_v2/rfid_solenoid_access.ino`

The firmware still uses a simple UID allowlist. Replace the placeholder UIDs with the real authorized card UIDs before deployment.

Key limitations

- UID-only access control is vulnerable to cloned cards.
- Powering a Nano from 12 V VIN is acceptable for a prototype but not ideal for thermal efficiency.
- A dedicated 5 V buck regulator and proper bidirectional level shifting would be preferable in a hardened build.
- No enclosure, tamper switch, door sensor, or safe egress logic is included.